

The Foundational Formalism of Maximally Helicity Violating Amplitudes

Chapter 48: Massless Particles and Spinor-Helicity

March 29, 2026

Chapter 42 is the intellectual core of the entire MHV research program. Every result in Chapters 43, 48, 60, and beyond builds on the identities introduced here. *Read carefully.*

Contents

1 Introduction: Why Spinor-Helicity?

In Chapter 20 we wrote a massless momentum as $p^\mu = (E, \mathbf{p})$ with $p^2 = 0$. In the two-component Weyl formalism this is encoded in a 2×2 matrix

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p^0 - p^3 \end{pmatrix}, \quad p^2 = \det(p_{\alpha\dot{\alpha}}) = 0. \quad (1)$$

The vanishing of the determinant is the *massless* condition; for $p^2 = 0$ the matrix has rank 1, which means it *factorizes*:

$$\boxed{p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}} \iff p^\mu \text{ Weyl map } \lambda \otimes \tilde{\lambda}. \quad (48.1)$$

The objects λ_α (undotted) and $\tilde{\lambda}_{\dot{\alpha}}$ (dotted) are commuting (bosonic) Weyl spinors. They are constrained only by the little-group scaling

$$\lambda_\alpha \rightarrow t \lambda_\alpha, \quad \tilde{\lambda}_{\dot{\alpha}} \rightarrow t^{-1} \tilde{\lambda}_{\dot{\alpha}}, \quad t \in \mathbb{C}^*. \quad (48.2)$$

Since $p_{\alpha\dot{\alpha}}$ is invariant, the individual spinors carry conformal weight. This is crucial for helicity.

2 Angle and Square Brackets

The fundamental Lorentz-invariant contractions are:

$$\langle ij \rangle \equiv \varepsilon^{\alpha\beta} \lambda_{i\alpha} \lambda_{j\beta}, \quad (48.16)$$

$$[ij] \equiv \varepsilon^{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_{i\dot{\alpha}} \tilde{\lambda}_{j\dot{\beta}}. \quad (48.17)$$

Both are antisymmetric: $\langle ij \rangle = -\langle ji \rangle$, $[ij] = -[ji]$, and $\langle ii \rangle = [ii] = 0$ (null-spinor property).

In Cadabra2:

```
cadabra2.AntiSymmetric(r"\bra{k}{p}") # angle <kp>
cadabra2.AntiSymmetric(r"\sbra{k}{p}") # square [kp]
cadabra2.AntiSymmetric(r"\epsilon_{\alpha\beta}")
cadabra2.AntiSymmetric(r"\epsilon^{\alpha\beta}")
```

Cadabra2 symbolic result for the angle bracket:

$$kp \quad (\text{undotted spinors, antisymmetric}). \quad (2)$$

Cadabra2 symbolic result for the square bracket:

$$k\bar{p} \quad (\text{dotted spinors, antisymmetric}). \quad (3)$$

3 Schouten Identity

For any three spinors $\lambda_i, \lambda_j, \lambda_k$:

$$ij\,kl + jk\,il + ki\,jl = 0. \quad (48.21)$$

This is the spinor analogue of the vector triple-product identity. Cadabra2 verification:

$$\text{Schouten} = (i, j)(k, l) + (j, k)(i, l) + (k, i)(j, l) = 0 \quad \checkmark. \quad (4)$$

4 Dot Products from Spinors: $2p_i \cdot p_j = \langle ij \rangle [ji]$

The key computational identity:

$$2p_i \cdot p_j = \langle ij \rangle [ji] \quad (48.17)$$

This is the workhorse: it replaces metric contractions $p_i \cdot p_j$ with bracket products. In the spinor-helicity formalism you never write $p_i \cdot p_j$ — you write $\langle ij \rangle [ji]$ instead.

Numerical check (affine coordinates, $\theta = 60^\circ$):

Quantity	Value
z_1 (affine param of leg 1)	0.0000
z_2 (affine param of leg 2)	-1.7321
$\langle 12 \rangle$	-1.7321
$[12]$	-1.7321
$s_{12} = \langle 12 \rangle [12]$	6.0000

5 Little Group Scaling and Helicity

Under $\lambda_i \rightarrow t_i \lambda_i$, $\tilde{\lambda}_i \rightarrow t_i^{-1} \tilde{\lambda}_i$:

- $\langle ij \rangle \rightarrow t_i t_j \langle ij \rangle$ (each λ contributes 1 t)
- $[ij] \rightarrow t_i^{-1} t_j^{-1} [ij]$ (each $\tilde{\lambda}$ contributes 1 t^{-1})
- $p_i \cdot p_j = \frac{1}{2} \langle ij \rangle [ji] \rightarrow t_i^2 p_i \cdot p_j$

A field of helicity h scales as t^{-2h} . For $h = +1$ (gluon): weight -2 ; for $h = -1$ (gluon negative helicity): weight $+2$.

For massless $k^\mu = (E, 0, 0, E)$ along $+z$:

$$u_+(k) \propto \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u_-(k) \propto \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (5)$$

Numpy verification ($E = 1.0, p_z = 0.5, \phi = 0.79$):

Quantity	Value
$u_+(E, p_z, \phi)$	$(1.225 + 0.000j, 0.500 + 0.500j)$
$u_-(E, p_z, \phi)$	$(-0.500 + 0.500j, 1.225 + 0.000j)$
$u_+^\dagger u_+$	$2.0000 = 2E$
$u_-^\dagger u_-$	$2.0000 = 2E$
$u_+^\dagger u_-$	$0.000000 + 0.000000j \approx 0$ (orthogonal)

6 Gluon Polarization Vectors

In axial gauge with reference momentum q :

$$\varepsilon_\mu^+(k; q) = \frac{\langle q | \gamma_\mu | k \rangle}{\sqrt{2} \langle qk \rangle}, \quad \varepsilon_\mu^-(k; q) = \frac{[q | \gamma_\mu | k \rangle}{\sqrt{2} [qk]}. \quad (48.\star)$$

Properties: $k^\mu \varepsilon_\mu^\pm = 0$ (transverse), $q^\mu \varepsilon_\mu^\pm = 0$ (reference gauge), $\varepsilon^\pm \cdot \varepsilon^\pm = -1$.

7 The Parke-Taylor Formula (MHV)

The crown jewel of spinor-helicity:

Little-group weight check: Under $\lambda_i \rightarrow t_i \lambda_i$: numerator $\langle jk \rangle^4 \rightarrow t_j^4 t_k^4 \langle jk \rangle^4$; each denominator $\langle ii+1 \rangle \rightarrow t_i t_{i+1} \langle ii+1 \rangle$. Net: $h = -1$ legs contribute t^4 (numerator) and each of their two adjacent denominators contribute t^{-1} , giving $t^{4-2} = t^2$ matching the required $t^{-2h} = t^2$ for $h = -1$. All positive-helicity legs give t^{-2} from their two adjacent denominators, matching $h = +1$. $\square$

8 Summary: The Spinor-Helicity Toolkit

Identity	Physical meaning
$p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$	Momentum = spinor outer product; rank-1 for massless
$2p_i \cdot p_j = \langle ij \rangle [ji]$	Mandelstam from brackets; no metric contractions
$\varepsilon_\mu^\pm = \langle q \gamma_\mu k \rangle / (\sqrt{2} \langle qk \rangle)$	Gluon polarization as spinor bilinear
Schouten: $\langle ij \rangle \langle kl \rangle + \langle jk \rangle \langle il \rangle + \langle ki \rangle \langle jl \rangle = 0$	Closure of the spinor algebra; replacement for $\delta^{(4)}$